

Detailed Changes and Verified Outputs

I prepared this detailed project update to present the completed invoice workflow, P90 benchmarking, historical evidence, batch processing, approval controls, vehicle classification and verified outputs.

PROJECT STATUS	DEMO DESIGN	P90 RULE	CAPTURE DATE
Verified	9 + 1 invoices	>10% and GBP 5	5 Aug 2026
End-to-end checks completed	Nine historical, one current	Both conditions required	Live local application state

Purpose	I prepared this document to explain what changed, what now works, and the exact screen outputs that can be presented to stakeholders.
Live evidence	Screens were captured from the running ClaimGuard application using invoice 9407 and the stored 10-invoice history.
Demo dataset	I also prepared a separate 10-PDF P90 demo set: the first two original invoices, eight controlled variants, and invoice 9510 as the current invoice.

1. Executive summary

The application now supports the complete journey from invoice intake to an explainable, auditable challenge decision.

1	2	3	4	5	6	7
Upload	Extract	Validate	Normalize	Benchmark	Decide	Audit
Single, multiple or folder PDF intake	Native text first; OCR provider optional	Line math, subtotal, VAT and total checks	Aliases map to standard repair items	Historical item prices produce P90	Accept, adjust, reject or challenge	Evidence, exports and immutable history

Outcome	What it means
Reliable invoice switching	Selecting a different uploaded invoice now refreshes its own extracted lines, totals, mappings, calculations and findings.
Explainable P90 decisions	Every evaluated line can display its current price, P90, difference, observation count, status and historical invoices used.
Controlled auto-approval	Extraction or mapping confidence above 95% is accepted automatically and shown with a green check; lower-confidence items remain reviewable.
Human decisions persist	Approve, adjust, reject and change-mapping actions update the stored record and appear in the audit trail.
Clean demo story	Nine invoices establish history. The tenth is evaluated without being included in its own benchmark.

2. Detailed change list

Each change reuses the existing workflow and data model wherever possible, so the product remains one connected system rather than a set of isolated features.

Area	Change	User-visible result	Evidence
Invoice intake	Added multiple-file and folder upload alongside single upload.	Every PDF is queued and retained separately; the invoice picker changes the active evidence view.	Browser-verified Documents screen and invoice selector.
Extraction	Kept native PDF extraction as the primary path; Azure Document Intelligence remains an optional configured provider.	Native-text invoices run without Tesseract. External keys can be added later without changing code.	All 10 demonstration PDFs parsed with OCR disabled.
Extraction confidence	Applied the >95% automatic acceptance rule.	High-confidence fields show a green Approved check instead of Pending or an unnecessary approval button.	Visible on Extracted Invoice.
Invoice calculations	Corrected summary-only labour, MOT VAT treatment, line extension and total reconciliation.	Synthetic invoice 9407 reports 18 passed, 1 not applicable and 0 failures.	Visible on Calculation Checks.
Ontology normalization	Expanded normalization and alias matching for differently written repair descriptions.	Oil Filter, OL Filter, Oil_Fil, Oil Filter Element and similar descriptions resolve to the same standard item.	Parser tests and live Repair Item Matching screen.
Mapping decisions	Reconnected Accept/Approve, Change, Link as bundle and Reject actions to the existing mapping review service.	Decisions persist and the active ontology bank remains authoritative.	Browser actions and audit records verified.
Benchmark database	Aggregated mapped invoice lines across stored invoices by canonical repair item and vehicle dimensions.	Benchmarks use combined reviewed history rather than only the selected invoice.	Benchmark dashboard shows observation counts and coverage.
P90 statistics	Added interpolated P90 beside Min, Max, Mean, Median, Mode and Count.	The benchmark number is visible and can be traced to exact invoice observations.	Dashboard and evidence drawer verified.
Current-invoice exclusion	Excluded the invoice being checked from its own line-item P90 calculation.	The decision is based only on earlier comparable observations.	Evidence drawer explicitly confirms exclusion.
Challenge rule	Centralized the shared threshold used by benchmark, findings and approval views.	Default rule: current price must be more than 10% above P90 and at least GBP 5.00 higher.	Threshold selector and all decision screens agree.
Challenge explanation	Added item, billed price, supported price, difference, percentage, source count and rationale.	The reviewer can see exactly what caused the challenge and by how much.	Review Findings and Challenge Decision verified.
Historical evidence	Restored detailed historical claim evidence and source links.	The drawer lists invoice number, date, original description and price used for the P90.	Nine historical rows shown in the live drawer.
Vehicle classification	Added static UK make/model lookup with normalization and aliases, including Vauxhall Adam Glam.	Vehicle category is carried into benchmarks instead of falling to Unclassified when a known alias exists.	Lookup import and classification tests passed.

Area	Change	User-visible result	Evidence
Audit and exports	Preserved append-only audit history and JSON, XLSX and SQLite exports.	Handler and automated decisions remain traceable and portable.	Audit & Reports screen verified.
User experience	Simplified the primary navigation and kept detailed tools in an Advanced section.	A first-time handler can follow Documents -> Benchmarks -> Findings -> Decision -> Summary.	Desktop and 1024 x 800 checks passed without overlap.
Demo dataset	Generated a controlled 10-PDF set with two originals and eight variants.	Description aliases exercise ontology mapping, while price variation creates presentation-ready challenge amounts.	Every PDF was parsed and calculation-checked.

3. P90 benchmark logic

P90 is calculated at normalized line-item level using earlier comparable invoice observations.

Decision rule	Challenge only when current price > P90 x 1.10 and current price - P90 >= GBP 5.00 .
Why two gates?	The percentage gate catches material relative increases; the GBP 5 floor prevents small differences such as 77 pence from becoming client-facing challenges.
History	The current invoice is excluded. After review it may contribute to future benchmarks under the stored-history policy.

9 + 1 invoice demonstration

Example item	Current invoice 9510	Historical P90	Difference	Expected decision
Oil filter	Controlled higher price	From the other nine PDFs	+GBP 10.81 (68.9%)	Challenge
Air filter	Controlled higher price	From the other nine PDFs	+GBP 25.82	Challenge
Pollen filter	Controlled higher price	From the other nine PDFs	+GBP 14.57	Challenge
Spark plugs	Controlled higher price	From the other nine PDFs	+GBP 40.62	Challenge
Engine oil	Controlled higher price	From the other nine PDFs	+GBP 48.96	Challenge
Full service labour	Controlled higher price	From the other nine PDFs	+GBP 193.80	Challenge
Disposal charge	Slightly above P90	From the other nine PDFs	+GBP 4.59	Within threshold - below GBP 5 floor

4. Verification completed

I verified the completed work at code, data-pipeline and browser levels before sharing it.

Check	Method	Result	Evidence
Frontend lint	npm run lint	PASS	No lint failures.
Frontend build	npm run build	PASS	Production bundle generated; only the standard large-chunk advisory remained.
Backend test suite	uv run pytest -q	PASS	Complete test suite passed; standard dependency deprecation warnings only.
Backend lint	uv run ruff check app tests	PASS	Python application and tests passed lint.
PDF extraction	PDFPipeline(ocr_enabled=False)	PASS	All 10 demonstration PDFs extracted using native text without Tesseract.
Calculation checks	Per-invoice validation	PASS	All 10 demo PDFs produced zero calculation failures.
Invoice switching	Browser user flow	PASS	Changing the invoice changed the displayed line items, prices and evidence.
P90 evidence	Browser user flow	PASS	Evidence drawer showed nine earlier invoices and confirmed current-invoice exclusion.
Decision actions	Browser user flow	PASS	Accept, adjust and reject controls responded; persisted actions appeared in the decision and audit views.
Responsive layout	1024 x 800 browser check	PASS	No overlapping table actions; horizontal overflow remains usable where needed.
Package integrity	unzip -t + SHA-256	PASS	Archive tested successfully with no secrets, caches or local database files.

5. Case overview

The handler sees one clear status: two lines need a decision, the proposed payable amount, the original invoice net and the total challenge amount.

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ClaimGuard

Motor claims review

Documents

Benchmarks

Review findings

Challenge decision

Summary

Case records

Audit trail

Advanced tools

API connected

VC Pilot Handler

Claims handler

Uploaded invoices

Switch invoice details; benchmarking still uses the combined reviewed dataset.

9407 · ST ALBANS CAR CLINIC (Synthetic Test Invoice)

Case overview

5 August 2026

Your case is ready for challenge approval. Review the 2 challenged invoice lines and decide.

Liability confirmed

Action needed

Documents processed

Complete

Findings ready

Complete

4

Challenge awaiting approval

Your decision needed

2 invoice lines need your decision

Review the challenged lines and approve the net challenge.

Review 2 challenged lines

Proposed payable net

£622.73

Proposed payable net (incl. £54.85 non-VAT MOT)

Original net

£640.32

Net invoice amount

Challenge amount

£17.59

6.14% of net invoice

Claim snapshot

Incident date

Third-party vehicle

YX68 HPA

Repairer

ST ALBANS CAR CLINIC

(Synthetic Test Invoice)

Invoice

9407

Why this is challenged

2 invoice lines are priced above our approved and historical prices. We challenge only when the positive variance is at least £5 and 5%.

View calculation evidence

All figures are shown exclusive of VAT unless stated.

ClaimGuard - Detailed changes and verified outputs

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6. Multiple and folder invoice upload

Single PDF, multiple PDF and folder inputs share the same processing workflow. The prior ontology and invoice banks are reused instead of re-uploaded for every file.

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Uploaded invoices

Switch invoice details; benchmarking still uses the combined reviewed dataset.

9407 · ST ALBANS CAR CLINIC (Synthetic Test Invoice)

Upload & Processing

Review pages

Draft analysis is available

PDF extraction and price analysis may continue regardless of liability. A challenge can only be issued later when liability is handler-confirmed ADMITTED or SPLIT LIABILITY.

Current repair invoice

READY

Upload a PDF and run the native-first pipeline.

Repair invoice PDF

Choose Files No file chosen

Synthetic_Invoice_8.pdf · 1 page available

Or choose a folder

Choose Files No file chosen

All PDF invoices in the selected folder are queued separately.

Upload and process

Ontology / reference bank

ACTIVE

Optional pilot or admin import

72 ontology items · active bank reused

Active bank reused

Previous invoice bank

READY

Optional supporting evidence

191 previous repair & service lines

Active bank reused

Processing run

Live status from the PDF pipeline for this case.

Stage	Method	Result	Status
File intake	SHA-256 + PDF signature validation	Synthetic_Invoice_8.pdf	PASS
Page classification	Native Text	1 page classified	PASS
Invoice extraction	Deterministic table and calculation pipeline	Existing extraction is available	PASS
Handler review	Audited page corrections	No page corrections	PASS

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7. Repair benchmark dashboard

The dashboard shows vehicle category, sample count, minimum, maximum, median and interpolated P90. The red count requires both the selected percentage threshold and the GBP 5 floor.

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9407 · ST ALBANS CAR CLINIC (Synthetic Test Invoice)

Repair benchmarks

Compare invoice repair costs with the governed history of mapped, invoice-only repair observations.

Vehicle category

All vehicle categories

Repair item

All repair items

Minimum sample

1+ observations

P90 alert threshold

More than 10% above P90

Aggregate repair benchmark summary

Combined stored invoice history. Red counts require both more than 10% above P90 and at least £5.00 difference; this descriptive dataset may include every stored invoice.

P90 · interpolated

Repair item	Vehicle category	Invoices	Min	Max	Median	P90	Above threshold	Invoices used
Engine oil (per litre)	Compact SUV	10	£29	£47	£41	£46	0	View 10
Brake pad wear sensor	Compact SUV	8	£12	£23	£16	£21	0	View 8
MOT test (Class IV)	Compact SUV	8	£55	£55	£55	£55	0	View 8
Oil & filter / environmental disposal charge	Compact SUV	8	£5	£8	£6	£7	0	View 8
Air filter (engine)	Compact SUV	7	£8	£16	£10	£16	0	View 7
Bulb (exterior, minor)	Compact SUV	6	£3	£28	£4	£16	1	View 6
Oil filter	Compact SUV	7	£7	£9	£8	£9	0	View 7
Screen wash	Compact SUV	7	£4	£4	£4	£4	0	View 7
Interim / small service labour	Compact SUV	6	£98	£136	£98	£128	0	View 6

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8. Exact P90 evidence

The evidence drawer lists the normalized item, current price, P90, difference, historical count and every invoice observation used. It also states that the current invoice is excluded.

ClaimGuard

Dashboard

Overview

Review Findings

Challenge Review

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Case Details

Full View

Challenge Review

Challenge Review

Oil & Filter Disposal

Current Price: £6.95

Challenge Price: £6.95

Challenge Count: 9

Challenge Status: Within threshold

Review Findings

Oil & Filter Disposal

Current Price: £6.95

Challenge Price: £6.95

Challenge Count: 9

Challenge Status: Within threshold

Oil & Filter Disposal

Persisted evidence used for this line comparison.

Normalized category: Oil & filter / environmental disposal charge

Current: £6.95, P90: £6.00, Difference: +£0.95, History used: 9

A challenge requires both more than 10% above P90 and at least £5.00 difference.

Interpolated percentile (PERCENTILE.INC). Current invoice excluded: yes.

Historical invoices used for P90

Invoice	Description	Price
90538 2025-09-26	Oil & Filter Disposal	£4.50
91283 2025-11-26	Oil & Filter Disposal	£4.50
9400 2025-12-10	Oil & Filter Disposal	£3.90
9401 2025-12-11	Oil & Filter Disposal	£4.25
9402 2025-12-12	Oil & Filter Disposal	£4.50
9403 2025-12-13	Oil & Filter Disposal	£4.95
9404 2025-12-14	Oil & Filter Disposal	£5.40
9405 2025-12-15	Oil & Filter Disposal	£5.90
9406 2025-12-16	Oil & Filter Disposal	£6.40

Mapping: 98%, Evidence: 65%, Challenge: 79/100

9. Explainable review findings

A challenged line shows the repairer charge, supported net price, challenge amount, evidence strength, approved ontology price and historical median. The P90 pilot appears above it.

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Motor claims review

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Uploaded invoices

Switch invoice details; benchmarking still uses the combined reviewed dataset.

9407 · ST ALBANS CAR CLINIC (Synthetic Test Invoice)

Review findings

1 of 2

APPROVED

Decide whether the evidence supports this price challenge.

Uploaded-invoice P90 benchmark pilot

90th percentile

The first extracted line is compared with the same normalized item on earlier invoices. The invoice being reviewed is excluded.

Normalized repair item	Current	P90 benchmark	Difference	Status
Oil & filter / environmental disposal charge 9 earlier invoices	£6.95	£6.00	+£0.95	Within threshold ⓘ

Invoice line

£9.79 high

Air Filter

1 each · Part

Repairer billed

£23.06

Supported net price

£13.27

Challenge amount

£9.79

Why it was flagged

Air Filter: billed £23.06; approved ontology £12.42 and historical median £13.27 from 9 comparable claims supports £13.27. Line challenge: £9.79.

Evidence used

View details

Approved price bank

£12.42

Air filter (engine)

Previous invoices

£13.27

Median of 9 comparable records

How the price was calculated

Ontology and history share a source lineage.

Decision context

The impact of accepting this finding.

Invoice

9407

Repairer

ST ALBANS CAR CLINIC (Synthetic Test Invoice)

Line reduction

£9.79

VAT impact

£1.96

Evidence strength

79/100

Review progress

0 decisions remaining.

1 / 2

Go to approval

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10. Final challenge decision

Reviewed lines are consolidated into the payable net, challenge amount, VAT impact and gross cash effect. Evidence remains available for each line.

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Uploaded invoices

Switch invoice details; benchmarking still uses the combined reviewed dataset.

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2 of 2

ready to issue

Approve challenge

Review every uploaded invoice before generating the combined challenge package.

Another invoice still needs review

13 findings remain across the uploaded invoices.

Review invoice 90538

Issuance is gated by liability

Confirm ADMITTED or SPLIT LIABILITY before generating the package.

Proposed payable net	Original net	Challenge amount	VAT impact	Gross cash effect	MOT
£622.73	£640.32	£17.59	£3.52	£21.11	£54.85
After the challenge amount					Outside VAT

The negotiation letter uses net figures; VAT and MOT are shown for context.

Liability confirmed

The invoice never decides fault

Policy claimguard-v1.4

60% ontology / 40% historic

Challenge findings

Item	Current net	Supported net	Challenge amount	Status	Evidence
Oil Filter	£15.80	£8.00	£7.80	Reviewed	View evidence
Air Filter	£23.06	£13.27	£9.79	Reviewed	View evidence

What will be generated

Challenge letter PDF

Professional negotiation letter with approved challenge figures and rationale.

Evidence appendix

Supporting evidence pack for the challenged line items.

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11. Extracted invoice and confidence

The selected invoice has its own metadata, vehicle category, totals and extracted line items. High-confidence rows display green Approved status instead of Pending.

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Unmatched items

Repair item library

API connected

VC Pilot Handler

Claims handler

Uploaded invoices

Switch invoice details; benchmarking still uses the combined reviewed dataset.

9407 · ST ALBANS CAR CLINIC (Synthetic Test Invoice)

View invoice

Run calculation checks

Extracted Invoice 9407

Review the raw extraction and correct any field before ontology mapping.

ST ALBANS CAR CLINIC (Synthetic Test Invoice)

Invoice number	9407	Invoice date	17 Dec 2025
Vehicle	BMW 320d Sport	Registration	YX68 HPA
Vehicle category	26-30 · High	Lookup match	alias
Mileage	0	Page range	1

Invoice totals

MOT is non-VAT and shown separately.

Parts net	£235.47
Labour net	£350.00
Taxable subtotal	£585.47
VAT	£117.09
MOT · non-VAT	£54.85
Gross invoice total	£757.41

Extracted line items

14 quantity-adjusted invoice lines · amounts shown net

Description	Part no.	Type	Qty	Unit	Unit price	Net total	VAT	Confidence	Status	
Oil & Filter Disposal	—	Part	1	job	£6.95	£6.95	20%	✓ Approved	APPROVED	
Oil Filter	—	Part	1	each	£15.80	£15.80	20%	✓ Approved	APPROVED	
Air Filter	—	Part	1	each	£23.06	£23.06	20%	✓ Approved	APPROVED	
Pollen Filter	—	Part	1	each	£12.12	£12.12	20%	✓ Approved	APPROVED	
Spark Plugs	—	Part	4	each	£9.21	£36.84	20%	✓ Approved	APPROVED	
Sundries - Grease/Oil Etc	—	Part	1	each	£4.50	£4.50	20%	✓ Approved	APPROVED	
Oil System Flush And Cleaner	—	Part	1	each	£8.50	£8.50	20%	✓ Approved	APPROVED	
Screen Wash	—	Part	1	each	£1.50	£1.50	20%	✓ Approved	APPROVED	

12. Calculation validation

Invoice 9407 shows 18 passed checks and one not-applicable labour detail check. MOT is explicitly excluded from VAT and totals reconcile.

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Uploaded invoices

Switch invoice details; benchmarking still uses the combined reviewed dataset.

9407 · ST ALBANS CAR CLINIC (Synthetic Test Invoice)

Calculation Checks

Deterministic arithmetic verifies line extensions, subtotals, VAT and the final invoice total.

Calculation checks

18 passed

1 not applicable

Taxable subtotal

£585.47

VAT verified

£117.09

Gross invoice

£757.41

MOT excluded from VAT

The £54.85 MOT charge is added after VAT and remains outside the negotiation VAT computation.

Validation ledger

Expected values are derived from extracted net line totals.

Check	Calculation
LABOUR TOTAL check	Labour is stated only in the invoice summary, so there are no detailed labour lines to reconcile. The stated labour amount is included in th
LINE MATH check	Quantity multiplied by net unit price for line 1.
LINE MATH check	Quantity multiplied by net unit price for line 2.
LINE MATH check	Quantity multiplied by net unit price for line 3.
LINE MATH check	Quantity multiplied by net unit price for line 4.
LINE MATH check	Quantity multiplied by net unit price for line 5.
LINE MATH check	Quantity multiplied by net unit price for line 6.
LINE MATH check	Quantity multiplied by net unit price for line 7.
LINE MATH check	Quantity multiplied by net unit price for line 8.
LINE MATH check	Quantity multiplied by net unit price for line 9.
LINE MATH check	Quantity multiplied by net unit price for line 10.
LINE MATH check	Quantity multiplied by net unit price for line 11.

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13. Repair item normalization

Thirteen items are approved and one remains reviewable. Each invoice description is linked to a standard repair item with its rationale and confidence.

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Switch invoice details; benchmarking still uses the combined reviewed dataset.

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Repair item matching

Check how each invoice line has been linked to a standard repair item.

Approved matches

13

Ready for price comparison

Awaiting review

1

A handler decision is still required

Average match confidence

92%

Based on the available invoice details

Bundle prices are never guessed

One invoice line can link to several repair items. If the price cannot be divided safely, it is sent for human review.

Review matches

13 approved · 1 awaiting review

Audit trail retained

Invoice item	Standard repair item	Link	Why this match	Match
Oil & Filter Disposal Part · 1 job	Oil & filter / environmental disposal charge		Oil & Filter Disposal: billed £6.95; approved ontology £5.25 (billed £1.70 higher) and historical median £8.00 from 9 comparable clai... ► Full explanation	✔
Oil Filter Part · 1 each	Oil filter	1:1	Oil Filter: billed £15.80; approved ontology £8.95 and historical median £7.75 from 10 comparable claims supports £8.00. Line... ► Full explanation	✔
Air Filter Part · 1 each	Air filter (engine)	1:1	Air Filter: billed £23.06; approved ontology £12.42 and historical median £13.27 from 9 comparable claims supports £13.27. Line... ► Full explanation	✔
Pollen Filter Part · 1 each	Cabin / pollen filter	1:1	Pollen Filter: billed £12.12; approved ontology £13.40 (billed £-1.28 higher) and historical median £10.95 from 4 comparable claims... ► Full explanation	✔
Spark Plugs Part · 4 each	Spark plug	1:1	Spark Plugs: billed £36.84; approved ontology £56.96 (billed £-20.12 higher) and historical median £56.92 from 2 comparable... ► Full explanation	✔
Sundries - Grease/Oil Etc Part · 1 each	Sundries - grease / oils	1:1	Sundries - Grease/Oil Etc: billed £4.50; the available governed evidence. The supported net price is £4.50. producing a £0.00...	✔

14. Source-document traceability

Ten documents and twelve pages remain individually traceable, with the extraction method, confidence, group, rotation and correction action visible.

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9407 · ST ALBANS CAR CLINIC (Synthetic Test Invoice)

Document Pages

Review live page images, machine classifications and handler corrections before extraction.

10 processed documents

12 pages · live pipeline evidence

Page	Classification	Source & confidence	Group	Rotation	Action
Page 1 Original_Invoice_1.pdf INVOICE pending Pipeline Native Text · 78% confidence invoice:90538 0° Correct					
Page 2 Original_Invoice_1.pdf INVOICE pending Pipeline Native Text · 78% confidence invoice:90538 0° Correct					
Page 1 Original_Invoice_2.pdf INVOICE pending Pipeline Native Text · 78% confidence invoice:91283 0° Correct					
Page 2 Original_Invoice_2.pdf INVOICE pending Pipeline Native Text · 78% confidence invoice:91283 0° Correct					
Page 1 Synthetic_Invoice_1.pdf INVOICE pending Pipeline Native Text · 78% confidence invoice:9400 0° Correct					
Page 1 Synthetic_Invoice_2.pdf INVOICE pending Pipeline Native Text · 78% confidence invoice:9401 0° Correct					
Page 1 Synthetic_Invoice_3.pdf INVOICE pending Pipeline Native Text · 78% confidence invoice:9402 0° Correct					
Page 1 Synthetic_Invoice_4.pdf INVOICE pending Pipeline Native Text · 78% confidence invoice:9403 0° Correct					

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15. Audit and portable outputs

The case can be exported as JSON, XLSX or SQLite. Human and automated decisions are retained in an append-only audit history.

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Audit & Reports

CG-2026-0048

Reproducible outputs stamped with policy claimguard-v1.4 and ontology ontology-v0-bootstrap.

JSON evidence pack

Machine-readable claim, extraction and challenge record

Download JSON

Excel review workbook

Line-level mappings, comparables and calculation checks

Download XLSX

SQLite case database

Portable pilot record and append-only audit trail

Download SQLITE

Audit history

3 append-only human and automated events

Time	Actor	Event	Value	Record hash
31/07/2026, 22:27:05	codex.qa	CLAIM CREATED	case	ac84f3ac37b95965c3ab8c5bbce3c508c...
05/08/2026, 22:34:51	pilot.handler	CHALLENGE APPROVED	approved	bee7db83a6457a6c579a5016c255c2e4b...
05/08/2026, 22:35:59	pilot.handler	CHALLENGE APPROVED	approved	c925b6c10e94c0f0b29247aadd38a2962...

Audit-ready by construction

Extraction, model suggestion, handler correction, policy calculation and report versions remain separate, traceable records.

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16. Recommended demonstration sequence

Use this simple story when presenting the application.

Step	What to say or show
1. Start with the problem	Repair invoices use inconsistent descriptions and prices. Manual comparison is slow and difficult to explain.
2. Upload the history	Upload the first nine PDFs from output/pdf/p90-demo-invoice-set. They become the earlier comparable observations.
3. Upload the current invoice	Upload Invoice_10_Variant.pdf, invoice 9510. It is the invoice being checked and is excluded from its own P90.
4. Show extraction	Open Extracted Invoice and point out that its own line descriptions, prices and totals are visible. High-confidence values are already approved.
5. Show normalization	Open Repair Item Matching and explain that spelling variations still map to the same standard repair item.
6. Show calculations	Open Invoice Calculations and confirm line math, VAT and invoice totals pass.
7. Show P90	Open Benchmarks. Select the 10% threshold, then explain that the benchmark comes only from earlier invoices.
8. Open the evidence	Use the information control to show the exact invoice numbers, descriptions and prices used to calculate P90.
9. Make the decision	Open Review Findings. Items above both thresholds are challenged; small differences under GBP 5 remain within threshold.
10. Finish with governance	Open Challenge Decision and Audit Trail to show the final amount, evidence package and recorded handler decisions.

17. Project files and operating notes

The project is ready to run on a client laptop, with external configuration kept separate from the code.

Item	Location / instruction
Application archive	ClaimGuard application ZIP
Quick start	Project setup guide included in the application ZIP
P90 demo PDFs	output/pdf/p90-demo-invoice-set/
Frontend setup	Copy .env.example to .env, install dependencies, then run npm run dev.
Backend setup	Copy backend/.env.example to backend/.env, run migrations and bootstrap, then start Uvicorn.
Native PDFs	No Tesseract installation is required for the supplied native-text PDF demonstration.
Azure Document Intelligence	Optional. Add the endpoint and key to backend/.env if scanned-image OCR is required later.
Gemini	Optional for configured language-model features. Add its key only in backend/.env; never place secrets in frontend source files.
Vehicle lookup expansion	Append make/model rows to sample-data/vehicle_category_lookup.csv and run the documented import command.
Archive checksum	28ab4a5f728d1889968bb563645196b1f81af7948852b531bce99e2afed5f170

Project conclusion	The application is ready to demonstrate with native PDF invoices. The P90 decision is visible, evidence-backed and consistent across the benchmark, findings and approval screens. External Azure or Gemini credentials remain optional deployment configuration and are not included in the archive.
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